

Effects of polyphosphate additives on the pH of processed chicken exudates and the survival of *Campylobacter*.



Campylobacter spp. are nutritionally fastidious organisms that are sensitive to normal atmospheric oxygen levels and lack homologues of common cold shock genes. At first glance, these bacteria seem ill equipped to persist within food products under processing and storage conditions; however, they survive in numbers sufficient to cause the largest number of foodborne bacterial disease annually. A mechanism proposed to play a role in *Campylobacter* survival is the addition of polyphosphate-containing marinades during poultry processing. *Campylobacter jejuni* and *Campylobacter coli* strains incubated in chicken exudates collected from poultry treated with a marinade demonstrated considerable survival advantages (1 to 4 log CFU/ml) over the same strains incubated in chicken exudate from untreated birds. Polyphosphates, which constitute a large portion of the commercial poultry marinades, were shown to account for a majority of the observed influence of the marinades on *Campylobacter* survival. When six different food grade polyphosphates (disodium pyrophosphate, tetrasodium pyrophosphate, pentasodium triphosphate, sodium polyphosphate, monosodium phosphate, and trisodium phosphate) were utilized to compare the survival of *Campylobacter* strains in chicken exudate, significant differences were observed with regard to *Campylobacter* survival between the different polyphosphates. It was then determined that the addition of polyphosphates to chicken exudate increased the pH of the exudate, with the more sodiated polyphosphates increasing the pH to a greater degree than the less sodiated polyphosphates. It was confirmed that the change in pH mediated by polyphosphates is responsible for the observed increases in *Campylobacter* survival.